PRACTICAL 1

Aim: Develop a simple “Hello World” Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

Software Name: Android Studio

Version: Panda (P): 2025.3

1. Copy the installer file from your Pen-drive and paste it onto your Computer's Desktop.
2. Run the Installer by Double-clicking the copied file.
3. Click **Next** to Continue.

4. Make sure **Android Virtual Device (AVD)**is checked and click **Next**.

5. Browse location and click **Next**.

6. If you want you can create shortcut or not and then click **Install**.

7. After successfully installing setup click **Next**.

8. As Android Studio has been installed and **check** the tick box to start Android Studio and click **Finish** to close the setup.

9. For allowing Google to collect usage data for Android Studio click **Don’t Send**.

10. As download starts wait for it to complete.

11. As the screen pops that shows Welcome we will start setting up the Android Studio Wizard by clicking **Next**.

12. For Installation type select **Standard Type** and click **Next**.

13. Review the list of SDK components to be downloaded and click **Next**.

14. For License Agreement **Accept**the Terms and Conditions and click **Next**.

15. Wait for Android Studio Setup Wizard to complete its download.

16. After downloading click **Finish**.

17. Launch the Android Studio and click onto **Projects** and select **New Project**.
18. Ignore the update that pops on the screen.

19. For Template select **Phone and Tablet**, then select **Empty Views Activity**and click **Next**.

20. For creating a new empty activity start by naming it **Pract1**, check the saving location, then select Language as **Java**, Minimum SDK as **API 24 ("Nougat";Android 7.0)**, BiuldConfifuration Language as **Kotlin DSL (build.gradle.kts)** and then click **Finish**.

21. Let for Gradle Project to complete its loading.
22. Click **Setup SDK**and then select **Oracle OpenJDK 26.0.1 C:\Java\jdk-26.0.1**.

23. In device manager select **Create Virtual Device**.

24. Select phone: **Pixel 8**.

25. Select the checkbox to **show system images with SDK Extensions** and click **Finish**.

26. Wait for SDK Component to get installed and after installing click **Finish**.

27. In **App > java > MainActivity.java**
28. Click **Run App** or **Shift+F10** to run the code.

```
package com.example.pract1;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });
  }
}
```

29. **A mobile screen will appear with the output.**

PRACTICAL 2

Aim: Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

1. **Go to res > values > colors.xml > add color (eg.blue)**

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<color name="black">#FF000000</color>
<color name="white">#FFFFFFFF</color>
<color name="blue">#038DCB</color>
</resources>
```

2. **Go to res > values > string.xml > string add classname TYIT**

```
<resources>
<string name="app_name">Pract 2</string>
<string name="classname">TYIT</string>
</resources>
```

3. **Go to res > values > Create dimen.xml file > add dimentions**

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<dimen name ="text_size">30dp</dimen>
</resources>
```

4. **Go to res > values > themes.xml > edit code**

```
<?xml version="1.0" encoding="utf-8" ?>
<resources>
<style name="Base.Theme.Pract2"
parent="Theme.MaterialComponents.Light.DarkActionBar">
</style>
<style name="Theme.Pract2" parent="Base.Theme.Pract2" />
</resources>
```

5. **Go to res > drawable > new drawable file > background.xml**

6. **Go to res > layout > activity_main.xml**

```
<?xml version="1.0" encoding="utf-
8"?><androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/an
droid"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<TextViewandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/classname"
android:textSize ="@dimen/text_size"
android:textColor="@color/blue"
app:layout_constraintBottom_toBottomOf="pare
nt"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

7. **Go to app > java > MainActivity.java**

```
package com.example.pract2;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity
; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {

    @Override     protected void
onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.ma
in), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

8. **A mobile screen will appear with the Output.**

PRACTICAL 3

Aim: Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

1. Go to app > java > MainActivity.java

```
package com.example.pract3;

import android.os.Bundle;
import android.util.Log;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

import org.jspecify.annotations.Nullable;

public class MainActivity extends AppCompatActivity {


    String TAG="ActivityLifecycle";


    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
Log.d(TAG,"onCreate() Method");
    }

    @Override
    protected void onStart() {
super.onStart();
Log.d(TAG,"onStart() Method");
    }
```

```
    @Override     protected
void onDestroy() {
super.onDestroy();
Log.d(TAG,"onDestory() Method");
    }

    @Override     protected
void onRestart() {
super.onRestart();
Log.d(TAG,"onRestart() Method");
    }

    @Override     protected
void onPause() {
super.onPause();
Log.d(TAG,"onPause() Method");
    }

    @Override
protected void onStop() {
super.onStop();
Log.d(TAG,"onStop() Method");
    }

    @Override     protected
void onResume() {
super.onResume();
Log.d(TAG,"onResume() Method");
    }
}
```

2. Go to res > layout > activity_main.xml

```
<?xml version="1.0" encoding="utf-
8"?><androidx.constraintlayout.widget.ConstraintLayoutxmlns:android=
"http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"     android:layout_width="match_parent"
android:layout_height="match_parent"     tools:context=".MainActivity">
```

```
<TextViewandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hello World!"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

3. **Connect Device:** Run the application on an Android Virtual Device (Emulator) or a physical USB-debugging-enabled device.

4. **Open Logcat:** Click the **Logcat** tab located at the bottom-left pane.

5. **Filter Results:** Type tag:ActivityLifecycle in the Logcat search bar to filter out system clutter and see only your custom lifecycle logs.

A mobile screen will appear with the Output.

PRACTICAL 4

Aim: Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

1. **Go to res > layout > activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

2. **Go to res > Right Click > New > Fragment(Blank) > fagement_demo**

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Demo Fragment"
        android:textSize="30dp"
        android:textStyle="bold" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fragment Lifecycle Demo"
        android:textSize="20dp"
        android:layout_marginTop="10dp" />
```

```
</LinearLayout>
```

3. Go to res > Right Click > Fragement.java

```
package com.example.p4;

import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

    String TAG = "FragmentLifecycle";

    @Override
    public void onAttach(@NonNull Context context) {
        super.onAttach(context);

        Log.d(TAG, "onAttach() method");
    }

    @Override
    public void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        Log.d(TAG, "onCreate() method");
    }

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {
```

```
    Log.d(TAG, "onCreateView() method");

    return inflater.inflate(
            R.layout.fragment_demo,
            container,
            false
    );
}

@Override
public void onViewCreated(@NonNull View view,
                          @Nullable Bundle savedInstanceState) {

    super.onViewCreated(view, savedInstanceState);

    Log.d(TAG, "onViewCreated() method");
}

@Override
public void onStart() {
    super.onStart();

    Log.d(TAG, "onStart() method");
}

@Override
public void onResume() {
    super.onResume();

    Log.d(TAG, "onResume() method");
}

@Override
public void onPause() {
    super.onPause();

    Log.d(TAG, "onPause() method");
}

@Override
```

```
    public void onStop() {
        super.onStop();

        Log.d(TAG, "onStop() method");
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();

        Log.d(TAG, "onDestroyView() method");
    }

    @Override
    public void onDestroy() {
        super.onDestroy();

        Log.d(TAG, "onDestroy() method");
    }

    @Override
    public void onDetach() {
        super.onDetach();

        Log.d(TAG, "onDetach() method");
    }
}
```

4. Go to app > java > MainActivity.java

```
package com.example.p4;

import android.os.Bundle;

import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    @Override
```

```
    protected void onCreate(@Nullable Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        DemoFragment fragment = new DemoFragment();

        FragmentTransaction transaction =
                getSupportFragmentManager()
                        .beginTransaction();

        transaction.replace(
                R.id.fragment_container,
                fragment
        );

        transaction.commit();
    }
}
```

5. **A mobile screen will appear with the Output.**

PRACTICAL 5

Aim: Develop an Android application implementing Fragment-based UI navigation between multiple screens.

1. **Go to res > Right Click > New > Fragment(Blank) >FirstFragment.java**

```
package com.example.pract5;

import android.annotation.SuppressLint;
import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

import
android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;

public class FirstFragment extends Fragment {
   @SuppressLint("MissingInflatedId")
   @Nullable
@Override
   public View onCreateView(@NonNull LayoutInflater inflater,
@Nullable
ViewGroup container,
                        @Nullable Bundle savedInstanceState) {
       View view=inflater.inflate(R.layout.fragment_first, container,false);
Button btnsecond=view.findViewById(R.id.btnsecond);
btnsecond.setOnClickListener(v -> {getParentFragmentManager()
.beginTransaction()
.replace(R.id.fragment_container,newSecondFragment())
.addToBackStack(null)
.commit();
       });
       return view;
```

```
    }
}
```

2. Go to res > Right Click > New > Fragment(Blank) >SecondFragment.java

```
package com.example.pract5;

import android.annotation.SuppressLint;
import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

import
android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;

public class FirstFragment extends Fragment {
    @SuppressLint("MissingInflatedId")
    @Nullable
@Override
    public View onCreateView(@NonNull LayoutInflater inflater,
@Nullable
ViewGroup container,
                             @Nullable Bundle savedInstanceState) {
        View view=inflater.inflate(R.layout.fragment_first, container,false);
Button btnsecond=view.findViewById(R.id.btnsecond);
btnsecond.setOnClickListener(v -> {getParentFragmentManager()
.beginTransaction()
.replace(R.id.fragment_container,newSecondFragment())
.addToBackStack(null)
.commit();
        });
        return view;
    }
}
```

3. Go to app > java > MainActivity.java

```
package com.example.pract5;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.annotation.Nullable; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   @Override     protected void onCreate(@Nullable Bundle
savedInstanceState) {  super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);         if
(savedInstanceState == null)
       {
getSupportFragmentManager()
.beginTransaction()
.replace(R.id.fragment_container,newFirstFragment())
.commit();
       }
   }
}
```

4. **Go to res > layout >fragment_first**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android"
><TextViewandroid:layout_width="wrap_content"
android:layout_height="wrap_content"         android:text="First
fragment"         android:textSize="30dp"
android:layout_centerInParent="true"         android:id="@+id/txtf"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Go to second fragment"
android:id="@+id/btnsecond"
```

```
android:layout_below="@+id/txtf"
android:layout_centerHorizontal="true"
android:textSize="24dp"/>
</RelativeLayout>
```

5. **Go to res > Right Click > New > Fragment(Blank) >Fragment_second**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android" >

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Second Fragment"
android:textSize="30dp"
android:layout_centerInParent="true"
android:id="@+id/txts"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="back to first fragment"
android:id="@+id/btnfirst"
android:layout_below="@+id/txts"
android:layout_centerHorizontal="true"
android:textSize="24dp"/>
</RelativeLayout>
```

PRACTICAL 6

Aim: Develop an Android application demonstrating communication between Activity and Fragment.

1. **Go to res > layout > activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/fragment_container">

</FrameLayout>
```

2. **Go to res > Right Click > New > Fragment(Blank) > fagement_demo**

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24dp"
        android:id="@+id/txtmsg"
        android:text="Message from Activity" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Message to Activity"
        android:textSize="20dp"
        android:id="@+id/btnsend" />
```

```
</LinearLayout>
```

3. **Go to app > java > MainActivity.java**

```
package com.example.p5;
import android.os.Bundle;
import android.widget.Toast;

import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
        implements DemoFragment.FragmentCommunication {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        DemoFragment fragment =
                DemoFragment.newInstance(
                        "Hello Fragment Message From Activity"
                );

        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit();
    }

    @Override
    public void onMessageFromFragment(String msg) {
        Toast.makeText(this, msg, Toast.LENGTH_LONG).show();
    }
}
```

4. **Go to res > Right Click > New > Fragment(Blank) >Demofragment.java**

```
package com.example.p5;

import android.content.Context;
import android.os.Bundle;
import android.view.LayoutInflater;
```

```
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

    TextView txtmsg;
    Button btnsend;
    String msg;

    FragmentCommunication communication;

    public interface FragmentCommunication {
        void onMessageFromFragment(String msg);
    }

    @Override
    public void onAttach(@NonNull Context context) {
        super.onAttach(context);
        communication = (FragmentCommunication) context;
    }

    public static DemoFragment newInstance(String msg) {

        DemoFragment fragment = new DemoFragment();

        Bundle bundle = new Bundle();
        bundle.putString("Message", msg);

        fragment.setArguments(bundle);

        return fragment;
    }

    @Nullable
    @Override
    public View onCreateView(
            @NonNull LayoutInflater inflater,
            @Nullable ViewGroup container,
```

```
        @Nullable Bundle savedInstanceState) {

    View view = inflater.inflate(
            R.layout.fragment_demo,
            container,
            false
    );

    txtmsg = view.findViewById(R.id.txtmsg);
    btnsend = view.findViewById(R.id.btnsend);

    if (getArguments() != null) {
        msg = getArguments().getString("Message");
        txtmsg.setText(msg);
    }

    btnsend.setOnClickListener(v -> {
        communication.onMessageFromFragment(
                "Hello Activity Message From Fragment"
        );
    });

    return view;
  }
}
```

PRACTICAL 7

Aim: Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
xmlns:android="http://schemas.android.com/apk/res/android" >

<TextViewandroid:layout_width="match_parent
"        android:layout_height="wrap_content"
android:text="Student Registraction"
android:textSize="30dp"
android:layout_gravity="center"/>

<EditTextandroid:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="30dp"
android:hint="Enter name"/>

<EditTextandroid:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="30dp"
android:hint="Enter Email"/>
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:gravity="center"
android:orientation="horizontal">
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"
android:textSize="26dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Reset"
android:textSize="26dp"
```

```
android:layout_marginStart="30dp"/>
</LinearLayout>
</LinearLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract7;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

    }
}
```

PRACTICAL 8

Aim: Develop an Android application using RelativeLayout to align UI components dynamically.

1. **Go to res > layout > activity_main.xml**

```
<RelativeLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android" >

<TextViewandroid:id="@+id/txttitle"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_centerHorizontal="true"
android:paddingTop="30dp"
android:text="Relative layout Demo"
android:textSize="30dp" />
<ImageViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:src="@mipmap/ic_launcher"
android:layout_below="@+id/txttitle"
android:id="@+id/img"
android:layout_centerHorizontal="true"/>
<EditTextandroid:layout_width="match_pare
nt"
android:layout_height="wrap_content"
android:layout_below="@+id/img"
android:textSize="24dp"
android:hint="Enter Name"
android:id="@+id/txtname"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@+id/txtname"
android:text="Submit"
android:textSize="24dp"
android:id="@+id/btnsub"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Reset"
android:textSize="24dp"
```

```
android:layout_marginLeft="100dp"
android:layout_below="@+id/txtname"
android:layout_toRightOf="@+id/btnsub"/>

</RelativeLayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract8;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

   }
}
```

PRACTICAL 9

Aim: Develop an Android application using Table Layout to design a structured Login/Registration Form.

1. **Go to res > layout > activity_main.xml**

```
<TableLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:id="@+id/main"     android:padding="20dp"
android:stretchColumns="1"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<TableRow>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:text="Register Form"
android:textStyle="bold"/>
</TableRow>
<TableRow><TextViewandroid:layout_wid
th="wrap_content"
android:layout_height="wrap_content"
android:text="Name"
android:textSize="20dp"/>
<EditText
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Name"/>
</TableRow>
<TableRow><TextViewandroid:layout_wi
dth="wrap_content"
android:layout_height="wrap_content"
android:text="Email"
android:textSize="20dp"/>
<EditTextandroid:layout_width="match_parent"
android:layout_height="wrap_content"
```

```
android:textSize="20dp"
android:hint="Enter Email"/>
</TableRow>

<TableRow><TextViewandroid:layout_wid
th="wrap_content"
android:layout_height="wrap_content"
android:text="Password"
android:textSize="20dp"/>
<EditTextandroid:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Password"/>
</TableRow>

<TableRow>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:text="Registration"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:text="Reset"/>

</TableRow>

</TableLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract9;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

    }
}
```

PRACTICAL 10

Aim: Develop an Android application using ScrollView to display long scrollable content.

1. **Go to res > layout > activity_main.xml**

```
<ScrollViewandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android" >

<LinearLayoutandroid:layout_width="match_parent
" android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="20dp">

<TextViewandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Registration form"
android:textSize="20dp"
android:layout_gravity="center"
android:textStyle="bold"/>
<EditText
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter First Name"/>
<EditText
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Middle Name"/>
<EditText
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"          android:hint="Enter Last
Name"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Email"/>
```

```
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Mobile No"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Address "/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter City "/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter State"/>
<EditText
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Country "/>
<EditText
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Skills"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Password"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter  Confrim Password"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Qualification"/>
<EditTextandroid:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"
android:hint="Enter Hobbies"/>
```

```
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Sumit"
android:textSize="20dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Reset"
android:textSize="20dp"/>

</LinearLayout>
</ScrollView>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract10;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

   }
}
```

PRCATICAL 11

Aim: Develop an Android application using AbsoluteLayout demonstrating fixed positioning.

1. **Go to res > layout > activity_main.xml**

```
<AbsoluteLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android">

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Login"
android:textSize="30dp"
android:textStyle="bold"
android:layout_x="150dp"
android:layout_y="30dp"/>
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Username:"        android:layout_x="40dp"
android:layout_y="70dp"        android:textSize="25dp"/>
<EditTextandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:hint="Enter Username"
android:textSize="25dp"
android:layout_y="60dp"
android:layout_x="160dp"/>
<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:text="Password:"
android:layout_x="40dp"
android:layout_y="130dp"
android:textSize="25dp"/>
<EditTextandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:hint="Enter Password"
android:textSize="25dp"
android:layout_y="120dp"
android:layout_x="160dp"/>
```

```
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:id="@+id/button"
android:layout_x="56dp"
android:layout_y="191dp"
android:textSize="25dp"
android:text="Login"/><Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:id="@+id/button1"
android:layout_x="190dp"
android:layout_y="191dp"
android:textSize="25dp"
android:text="Reset"/>

</AbsoluteLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract11;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

   }
}
```

PRCATICAL 12

Aim: Develop an Android application using FrameLayout to overlay UI elements.

1. **Go to res > layout > activity_main.xml**

```
<FrameLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:android="http://schemas.android.com/apk/res/android">
<ImageViewandroid:layout_width="match_parent
"        android:layout_height="match_parent"
android:src="@mipmap/ic_launcher"
android:scaleType="centerCrop"/>
<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:text="Welcome to framelayout"
android:textSize="30dp"
android:textStyle="bold"
android:textColor="#F40347"
android:layout_gravity="center"/>
<Button         android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Click Here"
android:textSize="25dp"
android:layout_gravity="bottom|center_horizontal"/>

</FrameLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract12;

import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
```

```
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

    }
}
```

PRCATICAL 13

Aim: Develop an Android application using ListView with ArrayAdapter and implement item click handling.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:id="@+id/main"
android:padding="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:id="@+id/txt"
android:text="Programming Language"
android:textSize="30dp"
android:layout_gravity="center"/>

<ListViewandroid:layout_width="match_parent
"        android:layout_height="match_parent"
android:id="@+id/listview"/>


</LinearLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract13;

import android.os.Bundle; import
android.view.View; import
android.widget.AdapterView;
import
android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity{

ListViewlistView;


String[]lang={"Java","C++","C","Kotlin","C#","PHP","Javascript","Python
"};


   @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

listView=findViewById(R.id.listview);

ArrayAdapter<String> adapter=new ArrayAdapter<>(this,
android.R.layout.simple_expandable_list_item_1,lang);
listView.setAdapter(adapter);   listView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
         @Override           public void onItemClick(AdapterView<?>
parent, View view, int position, long id) {
             String item=lang[position];
Toast.makeText(MainActivity.this,"Selected:
"+item,Toast.LENGTH_LONG).show();

         }
      });

   }
}
```

PRCATICAL 14

Aim: Develop an Android application using GridView with Custom Adapter and image selection handling.

1. **Download 4 Images**

2. **Go to res > drawable > paste those images[Rename if required]**

3. **Go to res > layout > activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:padding="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:layout_gravity="center"
android:text="Image Gallery"
android:textSize="30dp"
android:textStyle="bold"
android:paddingBottom="10dp"/>
<GridViewandroid:layout_width="match_parent
"        android:layout_height="match_parent"
android:numColumns="2"
android:verticalSpacing="10dp"
android:horizontalSpacing="10dp"
android:stretchMode="columnWidth"
android:id="@+id/gridview"/>
</LinearLayout>
```

4. **Go to app > java > MainActivity.java**

```
package com.example.practical14;
```

```
import android.os.Bundle; import
android.view.View; import
android.widget.AdapterView;
import android.widget.GridView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
GridViewgridView;
String[] imageNames ={"img1","img2","img3","img4"};
int[] images
={R.drawable.image1,R.drawable.image2,R.drawable.image3,R.drawa
ble.ima ge4};


   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
gridView=findViewById(R.id.gridview);
CustomAdapter adapter= new CustomAdapter(this,
imageNames,images);   gridView.setAdapter(adapter);
gridView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
         @Override          public void onItemClick(AdapterView<?>
parent, View view, int position, long id) {
Toast.makeText(MainActivity.this,"Select:"+imageNames[position],
Toast.LENGTH_LONG).show();
         }
      });
   }
}
```

5. Go to kotlin+java> Right click > New java class >CustomAdapter

```
package com.example.practical14;
```

```
import android.content.Context;
import
android.view.LayoutInflater;
import android.view.View; import
android.view.ViewGroup; import
android.widget.BaseAdapter;
import
android.widget.ImageView;
import android.widget.TextView;

public class CustomAdapter extends BaseAdapter {
    Context context;
String[] imagenames;
int [] images;
LayoutInflaterinflater;     public CustomAdapter (Context
context,String[] imagenames, int[] images)
    {
this.context=context;
this.imagenames=imagenames;
this.images=images;
inflater=LayoutInflater.from(context);
    }

    @Override     public int
getCount() {         return
imagenames.length;
    }

    @Override
    public Object getItem(int position) {
return imagenames[position];
    }

    @Override     public long
getItemId(int position) {        return
position;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup
parent) {
```

```
      View
view=inflater.inflate(R.layout.grid_item,parent,false);
ImageViewimageView=view.findViewById(R.id.imageview)
;   imageView.setImageResource(images[position]);
TextViewtextView =view.findViewById(R.id.textview);
textView.setText(imagenames[position]);       return view;
   }
}
```

6. Go to res > Right click > New >XML[Layout xml file] >grid_item

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:layout_gravity="center"     android:padding="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<ImageView
android:layout_width="120dp"
android:layout_height="120dp"
android:id="@+id/imageview"
android:scaleType="centerCrop"/>
<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:id="@+id/textview"
android:textSize="18dp"
android:padding="5dp"/>
</LinearLayout>
```

PRCATICAL 15

Aim: Develop an Android application implementing RecyclerView as a modern alternative to ListView.

1. **Recycle View**

2. **Gradle Scripts >build.gradle.kts[Module:app] > edit code**

   implementation(“android.recycleview:recycle.4.0)

3. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:padding="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:text="RecyclerView"
android:layout_gravity="center"
android:textSize="30dp"
android:textStyle="bold"/>
<androidx.recyclerview.widget.RecyclerViewandroid:layout_width="mat
ch_parent"         android:layout_height="match_parent"
android:id="@+id/recyclerview"/>
</LinearLayout>
```

4. **Go to res > Right click > New >XML[Layout xml file] >Item_layout**

```
<?xml version="1.0" encoding="utf-
8"?><TextViewandroid:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="20dp"     android:padding="15dp"
android:id="@+id/txtitem"
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
/>
```

5. Go to kotlin+java> Right click > New java class >MyAdapter

```
package com.example.p15;

import
android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class MyAdaptor extends
RecyclerView.Adapter<MyAdaptor.ViewHolder> {

String[] data;   public
MyAdaptor(String[] data)
   {
this.data=data;
   }

public  static class ViewHolder extends
RecyclerView.ViewHolder{  TextViewtextView;
 public ViewHolder(@NonNull View itemview)
       {
          super(itemview);
textView=itemview.findViewById(R.id.txtitem);     }
   }
   @NonNull     @Override     public
MyAdaptor.ViewHolderonCreateViewHolder(@NonNull ViewGroup
parent, int viewType) {
       View view=
LayoutInflater.from(parent.getContext()).inflate(R.layout.item_layout,par
ent,false);
return new ViewHolder(view);
```

```
    }

    @Override     public void onBindViewHolder(@NonNull
MyAdaptor.ViewHolder holder, int position) {
holder.textView.setText(data[position]);
holder.itemView.setOnClickListener(v ->{
Toast.makeText(v.getContext(),"Selected:
"+data[position],Toast.LENGTH_LONG).show();
        });
    }

    @Override     public int
getItemCount() {
return data.length;
    }
}
```

6. Go to app > java >MainActivity.java

```
package com.example.p15;
import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat; import
androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

public class MainActivity extends AppCompatActivity {
RecyclerViewrecyclerView;
String[]
lang={"JAVA","C","C++","C#","J#","PYTHON","KOTLIN","PHP","JAVA
SCRIPT"};     @Override     protected void
onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
recyclerView=findViewById(R.id.recyclerview);
recyclerView.setLayoutManager(new LinearLayoutManager(this));
MyAdaptor adaptor=new MyAdaptor(lang);
recyclerView.setAdapter(adaptor);
```

```
    }
}
```

MODULE 2

PRCATICAL 1

Aim: Develop an Android application implementing AppBar (Toolbar) and Options Menu.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:orientation="vertical"     android:id="@+id/main"
xmlns:android="http://schemas.android.com/apk/res/android"
>

<androidx.appcompat.widget.Toolbarandroid:l
ayout_width="match_parent"
android:layout_height="?attr/actionBarSize"
android:id="@+id/toolbar"
app:title="ToolBar Demo"
android:background="#090909"
app:titleTextColor="#E7729A"/>

<TextViewandroid:layout_width="wrap_content
"        android:layout_height="wrap_content"
android:textSize="30dp"
android:textStyle="bold"
android:layout_gravity="center"
android:text="AppBar and Option Menu"
android:padding="22dp"/>
</LinearLayout>
```

2. **Go to res > right click > New > Android Resourse>menu_main[type> menu]**

3. **Go to res > menu > Right click > new > Menu Resource File > menu_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item
android:title="Home"
android:id="@+id/home"/>
<item
android:title="Settings"
android:id="@+id/settings"/>
<item
android:title="About"
android:id="@+id/about"/>
<item
android:title="Logout"
android:id="@+id/logout"/>
</menu>
```

4. **Go to app > java > MainActivity.java**

```
package com.example.p16;

import
android.annotation.SuppressLint;
import android.os.Bundle; import
android.view.Menu; import
android.view.MenuItem; import
android.widget.Toast;

import androidx.annotation.NonNull;
import
androidx.appcompat.widget.Toolbar;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
```

```
    Toolbar toolbar;

    @SuppressLint("MissingInflatedId")
    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
toolbar=findViewById(R.id.toolbar);
setSupportActionBar(toolbar);
    }

    @Override     public
booleanonCreateOptionsMenu(Menu menu) {
getMenuInflater().inflate(R.menu.menu_main,menu);
return true;
    }

    @Override     public
booleanonOptionsItemSelected(@NonNull MenuItem item) {
int id=item.getItemId();   if(id==R.id.home)
        {
Toast.makeText(this, "Home Selected",
Toast.LENGTH_SHORT).show();
        } else if (id==R.id.settings) {
Toast.makeText(this, "Settings Selected",
Toast.LENGTH_SHORT).show();
        } else if (id==R.id.about) {
Toast.makeText(this, "About Selected",
Toast.LENGTH_SHORT).show();
        } else if (id==R.id.logout) {
Toast.makeText(this, "Logout Selected",
Toast.LENGTH_SHORT).show();
        }
        return true;
    }
}
```

PRACTICAL 2

Aim: Develop an Android application demonstrating AlertDialog and custom DialogFragment.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:gravity="center"
xmlns:android="http://schemas.android.com/apk/res/android" >

<Button
android:id="@+id/btnalert"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Show AlertDialog"
android:textSize="20dp"/>

<Button
android:id="@+id/btncustom"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Show Custom Dialog"
android:textSize="20dp"/>
</LinearLayout>
```

2. **Res>Right Click>New>XML>Layout xml file>dialog_custom.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:layout_height="match_parent">

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Custom DialogFragment"
android:textSize="20dp"
android:textStyle="bold"/>

<EditText
android:id="@+id/txtname"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Name"
android:textSize="20dp"
```

```
android:layout_marginTop="20dp"/>

</LinearLayout>
```

3. **Go to app > java > MainActivity.java**

```
package com.example.pract17;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Button btnalert,btncustom;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

btnalert=findViewById(R.id.btnalert);
btncustom=findViewById(R.id.btncustom);

btnalert.setOnClickListener(v -> {
AlertDialog.Builder builder=new AlertDialog.Builder(this);
builder.setTitle("Delete Files")
  .setMessage("Are you sure you want to delete?")
  .setPositiveButton("Yes",(dialog, which) ->
   {String name=txtname.getText().toString}

        );

        });
  }
}
```

4. **kotlin>Right Click>New>javaclass>customDialogFragement**

```
package com.example.pract17;

import android.app.AlertDialog;
import android.app.Dialog;
import android.os.Bundle;
import android.view.LayoutInflater;
```

```
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.core.app.DialogCompat;
import androidx.fragment.app.DialogFragment;

public class CustomDialogFragment extends DialogFragment {
EditTexttxtname;

    @NonNull
    @Override
    public Dialog onCreateDialog(@Nullable Bundle savedInstanceState) {
AlertDialog.Builder builder=new AlertDialog.Builder(getContext());
LayoutInflater inflater=requireActivity().getLayoutInflater();
        View view=inflater.inflate(R.layout.dialog_custom,null);
txtname=view.findViewById(R.id.txtname);
builder.setView(view)
                .setTitle("Custom Dialog")
                .setPositiveButton("Submit",(dialog, which) ->
                {String name=txtname.getText().toString();
Toast.makeText(getActivity(),"Entered: "+name, Toast.LENGTH_SHORT).show();})
                .setNegativeButton("Cancel",(dialog, which) -> {
dialog.dismiss();
                });

        return builder.create();
    }
}
```

PRACTICAL 3

Aim: Develop an Android application implementing contextual menus and popup menus.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:gravity="center"
android:padding="20dp"
xmlns:android="http://schemas.android.com/apk/res/android" >

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:id="@+id/txt"
android:text="Long Press Me"
android:textSize="24dp"
android:textStyle="bold"/>

<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:id="@+id/btn"
android:text="Show popup menu"
android:textSize="24dp"/>

</LinearLayout>
```

2. **Go to app > java > MainActivity.java**

```
package com.example.pract18;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
```

```
import android.widget.PopupMenu;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
TextView txt;
   Button btnpop;


   @SuppressLint("MissingInflatedId")
   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
       txt=findViewById(R.id.txt);
btnpop=findViewById(R.id.btn);

registerForContextMenu(txt);

btnpop.setOnClickListener(v -> {
PopupMenupopupMenu=new PopupMenu(MainActivity.this,btnpop);

popupMenu.getMenuInflater().inflate(R.menu.mymenu,popupMenu.getMen
u());

popupMenu.setOnMenuItemClickListener(item -> {
Toast.makeText(this,item.getTitle(), Toast.LENGTH_LONG).show();
               return true;
           });
popupMenu.show();
       });

   }

   @Override
```

```
    public void onCreateContextMenu(ContextMenu menu, View v,
ContextMenu.ContextMenuInfomenuInfo) {
super.onCreateContextMenu(menu, v, menuInfo);

menu.setHeaderTitle("Select Option");
getMenuInflater().inflate(R.menu.mymenu,menu);

    }

    @Override
    public booleanonContextItemSelected(@NonNull MenuItem item) {
Toast.makeText(this, item.getTitle(), Toast.LENGTH_SHORT).show();
        return true;
    }
}
```

3. **Go to res > right click > New > Android Resourse>menu_main[type> menu]**

4. **Go to res > menu > Right click > new > Menu Resource File > mymenu.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

<item
android:title="Edit"
android:id="@+id/menuedit"/>
<item
android:title="Delete"
android:id="@+id/menudelete"/>
<item
android:title="Share"
android:id="@+id/menushare"/>

</menu>
```

PRACTICAL 4

Aim: Develop an Android application implementing Explicit and Implicit Intents.

1. **Go to res > layout > activity_main.xml**

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:gravity="center"
xmlns:android="http://schemas.android.com/apk/res/android" >

<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Explicit Intent"
android:textSize="25dp"
android:id="@+id/btnexplicit"/>

<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Implicit Intent"
android:textSize="25dp"
android:id="@+id/btnimplicit"/>
</LinearLayout>
```

2. **kotlin>Right Click>New>Activity>empty Views activity>activity_second**

Activity_second_xml

```
<LinearLayoutandroid:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:gravity="center"
xmlns:android="http://schemas.android.com/apk/res/android" >

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
```

```
android:text="Welcome to Second Activity"
android:textSize="25dp"/>
</LinearLayout>
```

Activity_second_java

```
package com.example.pract19;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class activity_second extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_second);

    }
}
```

3. **Go to app > java > MainActivity.java**

```
package com.example.pract19;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Button btnex,btnimp;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

btnex=findViewById(R.id.btnexplicit);
btnimp=findViewById(R.id.btnimplicit);

btnex.setOnClickListener(v -> {
          Intent intent=new Intent(MainActivity.this, activity_second.class);
startActivity(intent);
       });

btnimp.setOnClickListener(v -> {
          Intent intent=new Intent(Intent.ACTION_VIEW,
Uri.parse("https://www.google.com"));
startActivity(intent);
       });


   }
}
```

PRACTICAL 5

Aim: Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).

1. Go to res > layout > activity_main.xml

```
<LinearLayout android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    xmlns:android="http://schemas.android.com/apk/res/android" >


    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me"
        android:textSize="20dp"
        android:id="@+id/btnclick"/>
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Long Press Me"
        android:textSize="20dp"
        android:id="@+id/btnlongclick"/>


</LinearLayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract20;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Button btnclick,btnlongclick;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        btnclick=findViewById(R.id.btnclick);
        btnlongclick=findViewById(R.id.btnlongclick);

        btnclick.setOnClickListener(v -> {
            Toast.makeText(this,"Button
Clicked",Toast.LENGTH_SHORT).show();
        });

        btnlongclick.setOnLongClickListener(v -> {
            Toast.makeText(this,"Button Long Press
Clicked",Toast.LENGTH_SHORT).show();
            return true;
        });


    }
}
```

PRACTICAL 6

Aim: Develop an Android application using Adapters to populate Spinner or ListView dynamically.

1. Go to res > layout > activity_main.xml

```
<LinearLayout android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    xmlns:android="http://schemas.android.com/apk/res/android">

    <Spinner
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/spinner"/>
    <ListView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:id="@+id/listview"/>
</LinearLayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract21;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Spinner;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
```

```
    Spinner spinner;
    ListView listView;
    String[] cities = {"Mumbai", "Pune", "Delhi", "Chennai", "Hydrabad"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        spinner = findViewById(R.id.spinner);
        listView = findViewById(R.id.listview);

        ArrayAdapter<String> spinnerAdapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_spinner_item,
                        cities);
        spinner.setAdapter(spinnerAdapter);
        ArrayAdapter<String> listAdapter=new ArrayAdapter<>(this,
                android.R.layout.simple_list_item_1,cities);
        listView.setAdapter(listAdapter);

        spinner.setOnItemSelectedListener(new
AdapterView.OnItemSelectedListener() {
            @Override
            public void onItemSelected(AdapterView<?> parent, View view, int
position, long id) {
                Toast.makeText(MainActivity.this, "Selected: " + cities[position],
                        Toast.LENGTH_LONG).show();
            }

            @Override
            public void onNothingSelected(AdapterView<?> parent) {

            }
        });

    }
```

Practical 7

Aim: Develop an Android application implementing Started and Bound Services.

1. Go to res > layout > activity_main.xml

```
<LinearLayout  android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:padding="20dp"
android:gravity="center"
   xmlns:android="http://schemas.android.com/apk/res/android">

   <Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"         android:text="Start
Service"
       android:id="@+id/btnstart"/>

   <Button

      android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"        android:text="Stop
Service"        android:layout_marginTop="20dp"
      android:id="@+id/btnstop"/>

   <Button        android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:textSize="20dp"        android:text="Bind
Service"        android:layout_marginTop="20dp"
android:id="@+id/btnbind"/>

</Linearlayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract22;
```

```
import android.content.ComponentName;
import android.content.Intent; import
android.content.ServiceConnection; import
android.os.Bundle; import android.os.IBinder;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   Button btnstart, btnstop, btnbind;     MyBoundService
myBoundService;
   boolean isBound = false;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);        btnstart =
findViewById(R.id.btnstart);        btnstop =
findViewById(R.id.btnstop);        btnbind =
findViewById(R.id.btnbind);

      btnstart.setOnClickListener(v -> {
         Intent intent = new Intent(this, MyStartedService.class);
startService(intent);
      });
      btnstop.setOnClickListener(v -> {
         Intent intent = new Intent(this, MyStartedService.class);
stopService(intent);
      });
      ServiceConnection serviceConnection = new ServiceConnection() {
@Override
         public void onServiceConnected(ComponentName name, IBinder service) {
```

```
            MyBoundService.LocalIBinder binder =
(MyBoundService.LocalIBinder) service;
myBoundService = binder.getService();
            isBound = true;
            Toast.makeText(MainActivity.this, myBoundService.getMessage(),
Toast.LENGTH_LONG).show();
        }

        @Override
        public void onServiceDisconnected(ComponentName name) {              isBound
= false;

        }
    };

    btnbind.setOnClickListener(v -> {
        Intent intent = new Intent(this, MyBoundService.class);
bindService(intent, serviceConnection, BIND_AUTO_CREATE);
    });
  }
}
```

3. kotlin>Right Click>New>javaclass> MyStartedService.java

```
        package com.example.pract22;

        import android.app.Service; import
        android.content.Intent; import
        android.os.IBinder; import
        android.widget.Toast;

        import androidx.annotation.Nullable;

        import java.security.Provider;

        public class MyStartedService extends Service {
            @Override
            public int onStartCommand(Intent intent, int flags, int startId) {
        Toast.makeText(this, "Started Services Running",
```

```
            Toast.LENGTH_LONG).show();
        return START_STICKY;
    }

    @Override     public void
onDestroy() {
super.onDestroy();
        Toast.makeText(this, "Started Services Stopped",
Toast.LENGTH_SHORT).show();

    }

    @Nullable     @Override
    public IBinder onBind(Intent intent) {
        return null;
    } }
```

4. kotlin>Right Click>New>javaclass> MyBoundService.java

```
package com.example.pract22;


import android.app.Service; import
android.content.Intent; import
android.os.Binder; import
android.os.IBinder;

import androidx.annotation.Nullable;

public class MyBoundService extends Service {     private final
IBinder binder=new LocalIBinder();     public class
LocalIBinder extends Binder{        MyBoundService
getService(){
            return MyBoundService.this;
        }
    }

    @Nullable     @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
```

```
    public String getMessage(){        return
"Bound Service Started";     }
}
```

Practical 8

Aim: Develop an Android application implementing Notifications using Notification Manager.

1. Go to res > layout > activity_main.xml

```
<LinearLayout android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:gravity="center"
   xmlns:android="http://schemas.android.com/apk/res/android" >

   <Button        android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Show Notification"
android:id="@+id/btnnotify"
       android:textSize="20dp"/>

</LinearLayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract23;

import android.app.Notification; import
android.app.NotificationChannel; import
android.app.NotificationManager; import
android.os.Build; import android.os.Bundle;
import android.widget.Button;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity; import
androidx.core.app.NotificationCompat; import
androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   Button btnnotify;
   String CHANNEL_ID="MyChannel";
```

```
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);        EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
btnnotify=findViewById(R.id.btnnotify);

      btnnotify.setOnClickListener(v -> {
         showNotification();

      });
   }
   public void  showNotification()
   {
      NotificationManager manager=(NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
      if (Build.VERSION.SDK_INT>=Build.VERSION_CODES.O)
      {
         NotificationChannel channel=new
NotificationChannel(CHANNEL_ID,"MyChannel",
               NotificationManager.IMPORTANCE_DEFAULT);
manager.createNotificationChannel(channel);
      }
      NotificationCompat.Builder builder=new
NotificationCompat.Builder(this,
            CHANNEL_ID).setSmallIcon(android.R.drawable.ic_dialog_info)
            .setContentTitle("Notification Demo ")
            .setContentText("Hello from android notification")
            .setAutoCancel(false);
      manager.notify(1,builder.build());
   } }
```

3. Manifests -> AndroidManifests.xml

```
 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">

    <uses-permission
    android:name="android.permission.POST_NOTIFICATIONS"> </uses-
    permission>
```

Practical 9

Aim: Develop an Android application implementing Static and Dynamic Broadcast Receivers.

1. Go to res > layout > activity_main.xml

```
LinearLayout android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"     android:gravity="center"
   xmlns:android="http://schemas.android.com/apk/res/android" >

   <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Broadcast Receiver Example"
android:textSize="30dp"/> </LinearLayout>
```

2. Go to app > java > MainActivity.java

```
package com.example.pract24;

import android.content.IntentFilter;
import android.os.Bundle;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat; import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   MyReceiver myReceiver=new MyReceiver();

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);        EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);        IntentFilter
filter = new IntentFilter();
      filter.addAction("android.intent.action.AIRPLANE_MODE");
registerReceiver(myReceiver, filter);
```

```
    }

    @Override     protected void
onDestroy() {
super.onDestroy();
        unregisterReceiver(myReceiver);
    }
}
```

3. Manifests -> AndroidManifests.xml

```
</intent-filter>
    </activity>

    <receiver android:name=".MyReceiver" android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.AIRPLANE_MODE"/>
        </intent-filter>
    </receiver>
</application>
```

4. kotlin>Right Click>New>javaclass> Myreceiver.java

```
  package com.example.pract24;

import android.content.BroadcastReceiver;
import android.content.Context; import
android.content.Intent; import
android.widget.Toast;

public class MyReceiver extends BroadcastReceiver {
    @Override     public void onReceive(Context context, Intent
intent) {
        Toast.makeText(context, "Broadcast Received",
Toast.LENGTH_SHORT).show();
    } }
```

Practical 10

Aim: Develop an Android application implementing SQLite database CRUD operations.

1. Go to res > layout > activity_main.xml <LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical" android:padding="25dp"
xmlns:android="http://schemas.android.com/apk/res/android" >

<EditText android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter your Name"
android:textSize="25dp"
android:id="@+id/txtname"/>
<Button android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Insert" android:textSize="25dp"
android:id="@+id/btninsert"/>

<Button android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="View Date"
android:textSize="25dp"
android:id="@+id/btnview"/>

<EditText android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter update Name"
android:textSize="25dp"
android:id="@+id/txtupdatename"/>
<Button
android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Update"
android:textSize="25dp" android:id="@+id/btnupdate"/>

<Button android:layout_width="match_parent"
android:layout_height="wrap_content"

```
      android:text="Delete"        android:textSize="25dp"
android:id="@+id/btndelete"/>
   <TextView        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textSize="25dp"        android:padding="20dp"
      android:id="@+id/txtresult"/> </LinearLayout>
```

2. Go to app > java > MainActivity.java package com.example.pract25;

```
import android.os.Bundle; import
android.widget.Button; import
android.widget.EditText; import
android.widget.TextView; import
android.widget.Toast;

import androidx.activity.EdgeToEdge; import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets; import
androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {     EditText
txtname,txtnewname;
   Button btninsert,btnupdate,btnview,btndelete;
   TextView txtresult;
   DBHelper dbHelper;

   @Override
   protected          void          onCreate(Bundle          savedInstanceState)          {
super.onCreate(savedInstanceState);                              EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);          txtname=findViewById(R.id.txtname);
txtnewname=findViewById(R.id.txtupdatename);
btninsert=findViewById(R.id.btninsert);       btnupdate=findViewById(R.id.btnupdate);
btnview=findViewById(R.id.btnview);            btndelete=findViewById(R.id.btndelete);
txtresult=findViewById(R.id.txtresult);       dbHelper=new DBHelper(this);

      btninsert.setOnClickListener(v -> {          String
name=txtname.getText().toString();
```

```
dbHelper.insertData(name);
Toast.makeText(this, "Data Inserted",
Toast.LENGTH_SHORT).show();
      });
      btnview.setOnClickListener(v -> {           String
data=dbHelper.viewData();
txtresult.setText(data);
      });
      btnupdate.setOnClickListener(v -> {           String
oldname=txtname.getText().toString();           String
newname=txtnewname.getText().toString();
dbHelper.updateData(oldname,newname);
         Toast.makeText(this, "Data Updated",
Toast.LENGTH_SHORT).show();
      });
      btndelete.setOnClickListener(v -> {
         String name=txtnewname.getText().toString();
dbHelper.deleteData(name);          Toast.makeText(this, "Data
Deleted",
Toast.LENGTH_SHORT).show();
      }); }
}
```

3. **kotlin>Right Click>New>javaclass> DBHelper.java**

```
package com.example.pract25;

import android.content.ContentValues; import
android.content.Context; import
android.database.Cursor; import
android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

   public DBHelper(Context context)
   {
      super(context,"StudentDB",null,1);
   }
```

```
    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student("+"name TEXT)");

    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

    }
    public void insertData(String name)
    {
        SQLiteDatabase db=this.getWritableDatabase();
ContentValues values=new ContentValues();
values.put("name",name);        db.insert("student",null,values);
    }
    public String viewData(){
        SQLiteDatabase db=this.getReadableDatabase();
        Cursor cursor=db.rawQuery("SELECT * FROM student",null);
        String data="";
        while (cursor.moveToNext())
        {
            data+= cursor.getString(0)+"\n";
        }
        return data;
    }
    public void updateData(String oldvalue,String newvalue)
    {
        SQLiteDatabase db=this.getWritableDatabase();        ContentValues
values=new ContentValues();        values.put("name",newvalue);
        db.update("student",values,"name=?",new String[]{oldvalue});
    }
    public void deleteData(String name)
    {
        SQLiteDatabase db=this.getWritableDatabase();
db.delete("student","name=?",new String[]{name});
     }

}
```